

SAMSUNG

Samsung Innovation Campus

| Artificial Intelligence Course

Together for Tomorrow!
Enabling People

Education for Future Generations

Chapter 11.

AI Capstone Project Tutorial

AI Course



Unit 1.

Using a Ready-Made CNN Model

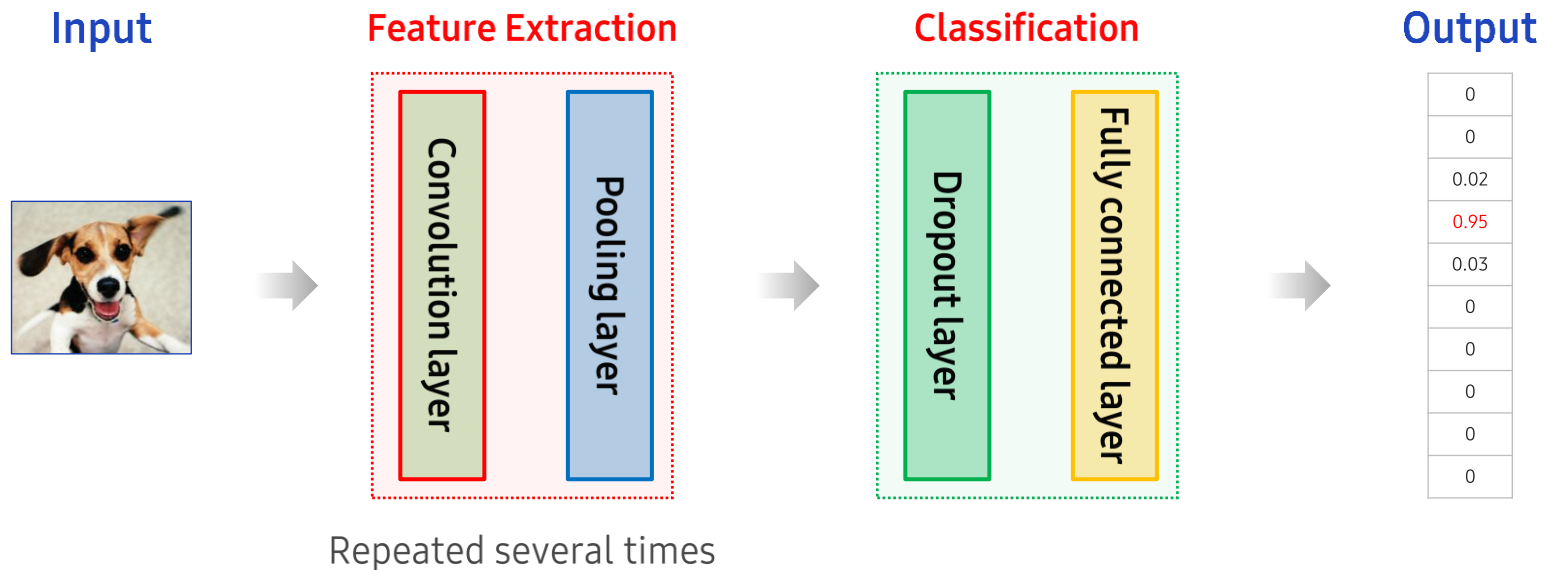
Using a Ready-Made CNN Model

| Why use a ready-made model?

- ▶ Building and training a deep CNN model from scratch can be quite costly (in time).
- ▶ There are deep CNN models that were optimized and trained on a large dataset (*).
- ▶ Reusing these ready-made models (**) to recognize new image classes can be more convenient.
- ▶ We will reuse only the convolutional base that is effective in extracting generic features.
- ▶ A new classification (fully connected dense) layer can be attached to adapt to the new dataset.
 - a) If the weights of the pre-trained convolution layers remain **frozen**, it's called "fine-tuning."
 - b) We can also update all the weights at once.

Using a Ready-Made CNN Model

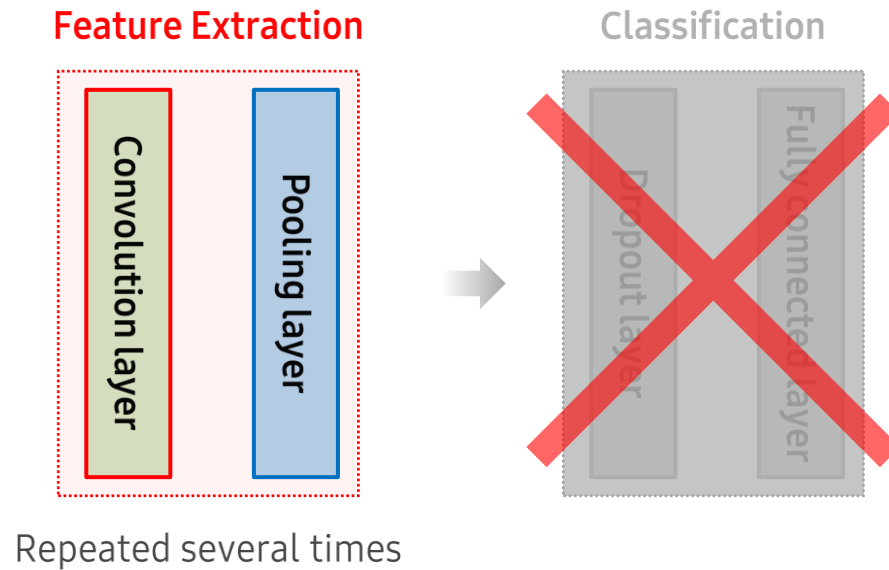
| Fine-tuning a pre-trained model



1) Suppose a deep learning CNN model trained on a dataset.

Using a Ready-Made CNN Model

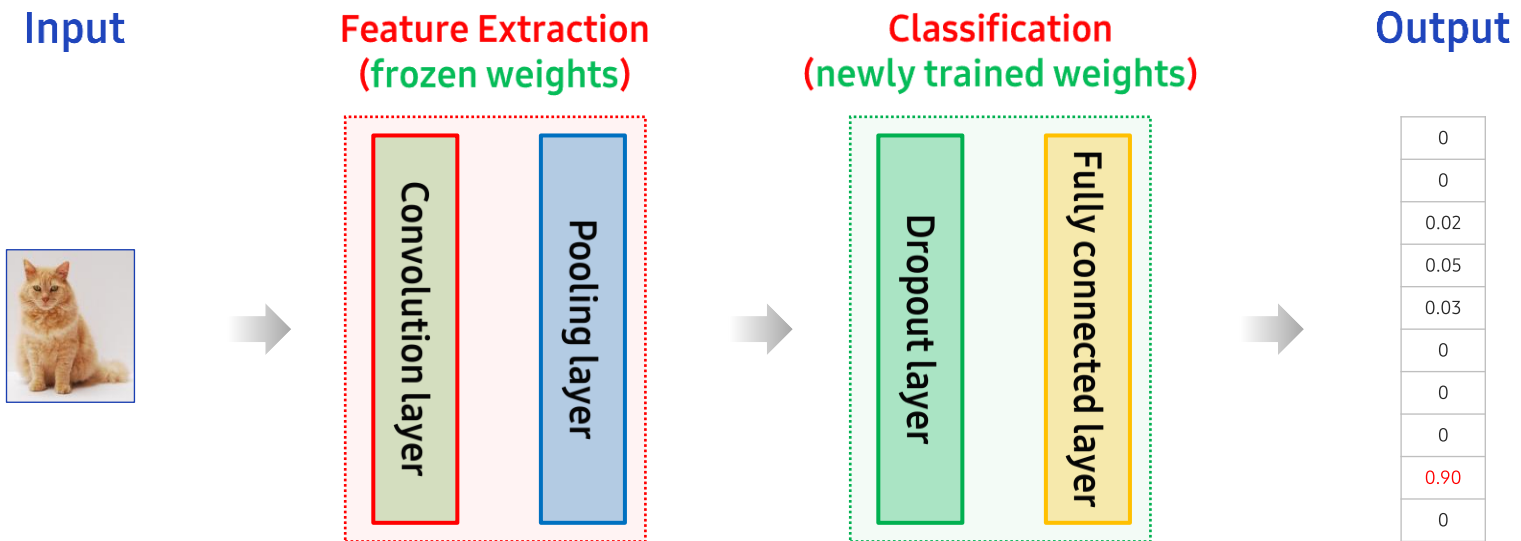
| Fine-tuning a pre-trained model



2) Remove the classification layer.

Using a Ready-Made CNN Model

| Fine-tuning a pre-trained model

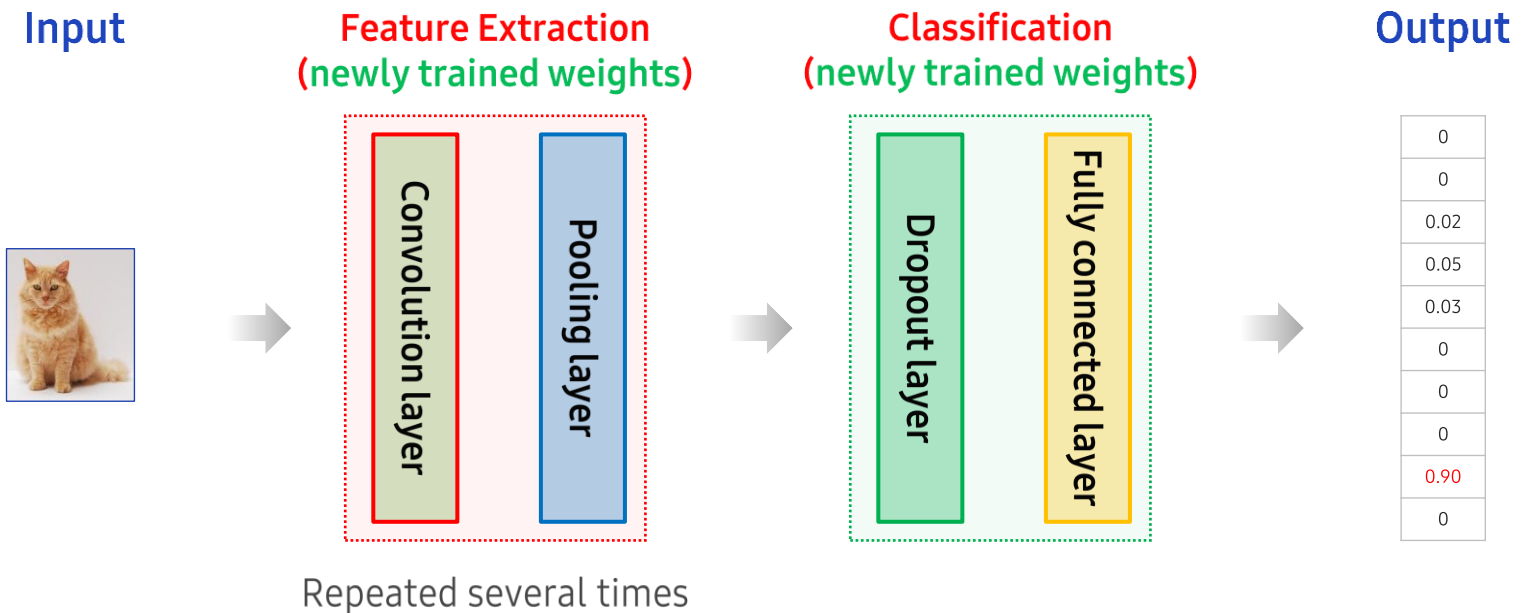


Repeated several times

3) Train the new classification layer with a new dataset.

Using a Ready-Made CNN Model

Updating all the weights at once



4) Train the whole network all over.

Using a Ready-Made CNN Model

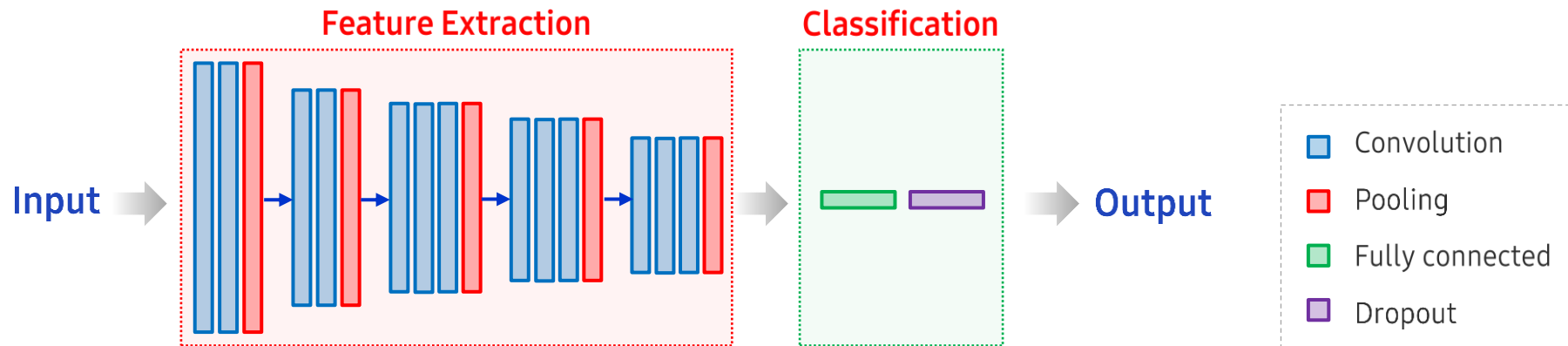
| Ready-made/Pre-trained models of Keras

Model Name	Size (MB)	Accuracy (top-1)	Parameters (in millions)
Xception	88	0.790	23
VGG16	528	0.713	138
VGG19	549	0.713	144
ResNet, ResNetV2	98 ~ 232	0.749 ~ 0.780	26 ~ 60
InceptionV3	92	0.779	24
InceptionResNetV2	215	0.803	56
MobileNet	16	0.703	4
MobileNetV2	14	0.713	4
DenseNet	33 ~ 80	0.750 ~ 0.773	8 ~ 20
NASNet	23 ~ 343	0.744 ~ 0.825	5 ~ 89

Using a Ready-Made CNN Model

| A well known pre-trained model: **VGG16 and VGG19**

- ▶ Developed in 2014. Runner-up of the 2014 ILSVRC competition
- ▶ The default input size: 224×224
- ▶ 3×3 convolution layers and 2×2 pooling layers
- ▶ Quite a large number of parameters
- ▶ Known to work well with fine tuning



Using a Ready-Made CNN Model

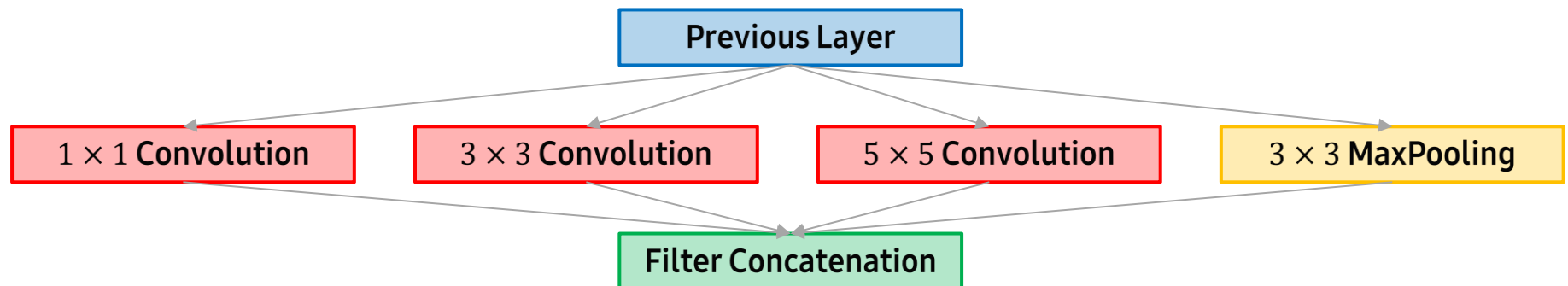
| A well known pre-trained model: **ResNet**

- ▶ Developed in 2015. Winner of the 2015 ILSVRC competition
- ▶ Shallower network that implements “residual learning”
- ▶ Non-linear layering structure with “shortcuts”
- ▶ The default input size: 224×224
- ▶ With fewer parameters than VGG achieves a higher accuracy

Using a Ready-Made CNN Model

| A well known pre-trained model: **InceptionV3**

- ▶ InceptionV1 was developed in 2014. Winner of the 2014 ILSVRC competition
- ▶ InceptionV3 is a further improved version.
- ▶ Uses multiple convolution filters (different kernel sizes) concatenated in parallel
- ▶ The default input size: 299×299
- ▶ With fewer parameters than VGG achieves a higher accuracy



Using a Ready-Made CNN Model

| How to further increase the accuracy in image classification

1) Make sure you have a large enough training set.

⇒ If not, augment the training dataset by rotation, cropping, resizing, blurring, etc.

2) Build an ensemble or stacking model.

⇒ Combine different CNN architectures and make them “**vote**.”

Unit 2.

AI Services and Pre-Trained AI Models

| 2.1. AI-based Services

| 2.2. Pre-Trained AI Models

Pre-Trained Large Language Models

| ChatGPT (Generative Pre-trained Transformer)

- ▶ <https://chat.openai.com/>
- ▶ [Transformer](#) Decoder based Generative Language Model
- ▶ Released in Winter of 2022 (~175 billion parameters)
- ▶ The time taken to complete 1 million users is only 5 days.
- ▶ Trained with large text corpus and human instruction tuning
- ▶ Text generation, sentiment analysis, document classification, document summary, keyword extraction, translation, Q&A, grammar correction, programming code creation, table creation, etc.
- ▶ GPT4 can even handle multimodals like image, voice, etc.
- ▶ ChatGPT service is available using [Open API](#).

tell me about chatgpt



ChatGPT is an AI language model developed by OpenAI. It is based on the GPT-3.5 architecture, which stands for "Generative Pre-trained Transformer 3.5." ChatGPT is designed to engage in natural language conversations with users, providing relevant and coherent responses to a wide range of prompts and queries.

GPT-3.5 is a highly advanced model that has been trained on a massive amount of text data from the internet, encompassing a wide variety of topics and writing styles. It has a knowledge cutoff, which means it was last trained on data up until September 2021, so it may not be aware of events or information that occurred after that time.

The model uses a transformer architecture, which is a type of deep learning model specifically designed for processing sequential data, such as text. Transformers have revolutionized the field of natural language processing (NLP) by enabling better language understanding and generation.

ChatGPT is capable of understanding and generating human-like responses. It can answer questions, provide explanations, engage in conversations, offer suggestions, and assist with a wide range of tasks. However, it's important to note that ChatGPT is an AI model and does not possess genuine understanding or consciousness. Its responses are generated based on patterns and probabilities learned during training.

OpenAI has developed ChatGPT for various applications, including customer support, content generation, language translation, and general-purpose conversational AI. It is designed to be adaptable and can be fine-tuned for specific tasks or domains, allowing developers to customize its behavior and improve its performance in specific contexts.

It's worth noting that while ChatGPT is highly advanced and can provide helpful information, it may occasionally generate incorrect or nonsensical responses. Users should exercise critical thinking and fact-check information obtained from AI models like ChatGPT to ensure accuracy and reliability.

Pre-Trained Large Language Models

| Bard

- ▶ <https://bard.google.com/>
- ▶ A Google service to counter the market expansion of ChatGPT
- ▶ Similar model architecture as ChatGPT
- ▶ Released in Spring of 2023 (~530 billion parameters)
- ▶ Similar function as ChatGPT
- ▶ Bard service is available using [Open API](#) (non-official).

tell me about bard

다른 답안 보기

Bard is a large language model chatbot developed by Google AI. It is based on the LaMDA family of large language models and later the PaLM LLM. Bard was first released in March 2023 and is currently available in English, Japanese, and Korean.

Bard can generate text, translate languages, write different kinds of creative content, and answer your questions in an informative way. It is still under development, but it has learned to perform many kinds of tasks, including:

- Following instructions and completing requests thoughtfully.
- Answering questions in a comprehensive and informative way, even if they are open ended, challenging, or strange.
- Generating different creative text formats of text content, like poems, code, scripts, musical pieces, email, letters, etc.

Bard is still under development, and it is not always perfect. It may sometimes give inaccurate or inappropriate responses. However, it is constantly learning and improving, and it is already a powerful tool that can be used for a variety of purposes.

Here are some of the things that Bard can do:

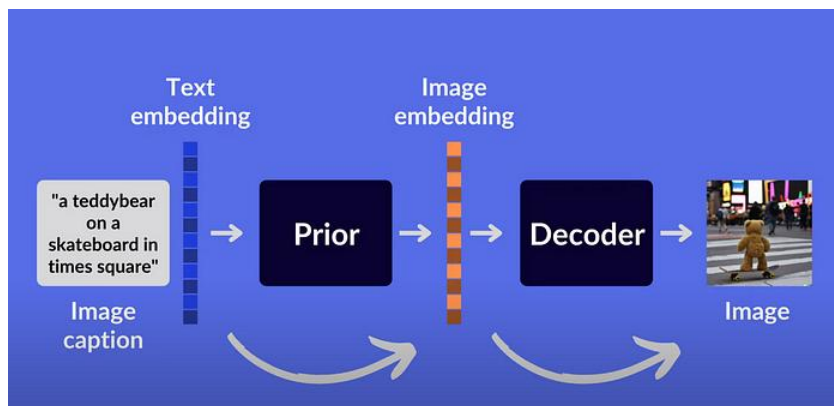
- **Generate text:** Bard can generate text in a variety of styles, including news articles, poems, code, scripts, musical pieces, email, letters, etc. It can also translate languages, write different kinds of creative content, and answer your questions in an informative way.
- **Access and process information:** Bard can access and process information from the real world through Google Search and keep its response consistent with search results.
- **Learn and adapt:** Bard is constantly learning and adapting. It can learn from the data it is trained on, as well as from the interactions it has with users.
- **Be creative:** Bard can be creative and generate new ideas. It can write poems, scripts, musical pieces, and other creative content.

Bard is a powerful tool that can be used for a variety of purposes. It can be used for education, entertainment, research, and more. As it continues to develop, Bard will become even more powerful and useful.

AI Based Image Generation Services

DALL-E 2

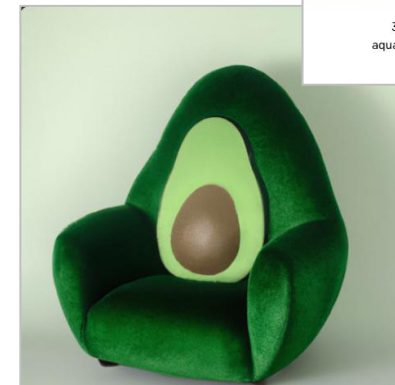
- ▶ <https://openai.com/dall-e-2>
- ▶ Transformer based image generation AI
- ▶ Insert a description of the image want to create; DALL-E 2 creates images.
- ▶ Various images, such as thumbnails and illustrations, can be created.
- ▶ **Outpainting**: Extend one's creativity by continuing an image beyond its original borders.



Source: [AssemblyAI](#)



3D render of a cute tropical fish in an aquarium on a dark blue background, digital art



An armchair in the shape of an avocado

AI Based Image Generation Services

Midjourney

- ▶ <https://www.midjourney.com/>
- ▶ Similar function as DALL-E 2
- ▶ Version 5 can produce higher quality output images.

Text Prompt (user input)

illustration style, a mid age asian woman with exhausted and annoying face, in shorts and slippers, both hands carrying grocery in plastic bags, walking out from a crowded super market in Taipei, it's summer, the weather is hot - @soupfang (fast) (수정됨)

Generated Image
(AI output)





Unit 2.

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| 2.2. Pre-Trained AI Models

Pre-Trained AI Model Repository

HuggingFace

- ▶ <https://huggingface.co/>
- ▶ Various pretrained models can be easily used using the Hugging Face. <https://github.com/huggingface/transformers#model-architectures>
- ▶ Multi AI Framework (Tensorflow, Pytorch, JAX/Flax) support.
- ▶ Easily download AI models and train them with domain-specific data.

```
[1] # install hugging face library
    !pip install transformers

[2] from transformers import AutoTokenizer, TFAutoModel

[3] # load bert base uncased tokenizer & model
    tokenizer = AutoTokenizer.from_pretrained('bert-base-uncased')
    model = TFAutoModel.from_pretrained('bert-base-uncased')

[4] inputs = tokenizer("Hello world!", return_tensors="tf")
    outputs = model(**inputs)
```

Model architectures

All the [model checkpoints](#) provided by 🤗 Transformers are seamlessly integrated from the [huggingface.co model hub](#) where they are uploaded directly by [users](#) and [organizations](#).

Current number of checkpoints: models 250,471

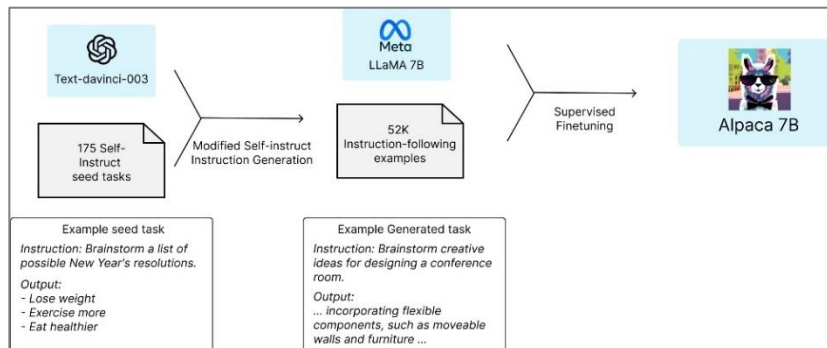
🤗 Transformers currently provides the following architectures (see [here](#) for a high-level summary of each them):

1. **ALBERT** (from Google Research and the Toyota Technological Institute at Chicago) released with the paper [ALBERT: A Lite BERT for Self-supervised Learning of Language Representations](#), by Zhenzhong Lan, Mingda Chen, Sebastian Goodman, Kevin Gimpel, Piyush Sharma, Radu Soricut.
2. **ALIGN** (from Google Research) released with the paper [Scaling Up Visual and Vision-Language Representation Learning With Noisy Text Supervision](#) by Chao Jia, Yinfei Yang, Ye Xia, Yi-Ting Chen, Zarana Parekh, Hieu Pham, Quoc V. Le, Yunhsuan Sung, Zhen Li, Tom Duerig.
3. **ARCLIP** (from BAAI) released with the paper [ARCLIP: Altering the Language Encoder in CLIP for Extended Language Capabilities](#) by Chen, Zhongzhi and Liu, Guang and Zhang, Bo-Wen and Ye, Fulong and Yang, Qinghong and Wu, Ledell.
4. **Audio Spectrogram Transformer** (from MIT) released with the paper [AST: Audio Spectrogram Transformer](#) by Yuan Gong, Yu-An Chung, James Glass.
5. **Autoformer** (from Tsinghua University) released with the paper [Autoformer: Decomposition Transformers with Auto-Correlation for Long-Term Series Forecasting](#) by Haixu Wu, Jiehui Xu, Jianmin Wang, Mingsheng Long.
.....
207. **XLS-R** (from Facebook AI) released with the paper [XLS-R: Self-supervised Cross-lingual Speech Representation Learning at Scale](#) by Arun Babu, Changhan Wang, Andros Tjandra, Kushal Lakhotia, Qiantong Xu, Naman Goyal, Kritika Singh, Patrick von Platen, Yatharth Saraf, Juan Pino, Alexei Baevski, Alexis Conneau, Michael Auli.
208. **XLSR-Wav2Vec2** (from Facebook AI) released with the paper [Unsupervised Cross-Lingual Representation Learning For Speech Recognition](#) by Alexis Conneau, Alexei Baevski, Ronan Collobert, Abdelrahman Mohamed, Michael Auli.
209. **YOLOS** (from Huazhong University of Science & Technology) released with the paper [You Only Look at One Sequence: Rethinking Transformer in Vision through Object Detection](#) by Yuxin Fang, Bencheng Liao, Xinggang Wang, Jiemin Fang, Jiyang Qi, Rui Wu, Jianwei Niu, Wenyu Liu.
210. **YOLO** (from the University of Wisconsin - Madison) released with the paper [You Only Sample \(Almost\) Once: Linear Cost Self-Attention Via Bernoulli Sampling](#) by Zhanpeng Zeng, Yunsang Xiong, Sathya N. Ravi, Shailesh Acharya, Glenn Fung, Vikas Singh.
211. Want to contribute a new model? We have added a [detailed guide and templates](#) to guide you in the process of adding a new model. You can find them in the [templates](#) folder of the repository. Be sure to check the [contributing guidelines](#) and contact the maintainers or open an issue to collect feedbacks before starting your PR.

Large Language Model

LLaMA & Alpaca

- ▶ <https://ai.facebook.com/blog/large-language-model-llama-meta-ai/>
- ▶ Large Language Model as ChatGPT, Bard
- ▶ Smaller parameters (6.7B ~ 65B) than GPT(175B) but similar performance
- ▶ LLaMA (6.7B) can run on a single GPU (v100).
- ▶ Alpaca (A Strong, Replicable Instruction-Following Model)
 - <https://crfm.stanford.edu/2023/03/13/alpaca.html>
 - Alpaca is a instruction tuning version of LLaMA 6.7B.



```
[1] !pip install -q -U bitsandbytes
!pip install -q -U git+https://github.com/huggingface/transformers.git
!pip install -q -U git+https://github.com/huggingface/peft.git
!pip install -q -U git+https://github.com/huggingface/accelerate.git
!pip install einops
!pip install tokenizers==0.13.3
!pip install sentencepiece

[2] import torch
from transformers import AutoTokenizer, AutoModelForCausalLM, BitsAndBytesConfig
from peft import PeftModel, PeftConfig

[3] # peft_model_list: https://huggingface.co/models?library=peft
peft_model_id = "aldrinc/llama-med-13b"
config = PeftConfig.from_pretrained(peft_model_id)
bnb_config = BitsAndBytesConfig(
    load_in_4bit=True,
    bnb_4bit_use_double_quant=True,
    bnb_4bit_quant_type="nf4",
    bnb_4bit_compute_dtype=torch.bfloat16
)
model = AutoModelForCausalLM.from_pretrained(config.base_model_name_or_path,
                                             quantization_config=bnb_config,
                                             device_map={"":0})
model = PeftModel.from_pretrained(model, peft_model_id)
tokenizer = AutoTokenizer.from_pretrained(config.base_model_name_or_path)

model.eval()

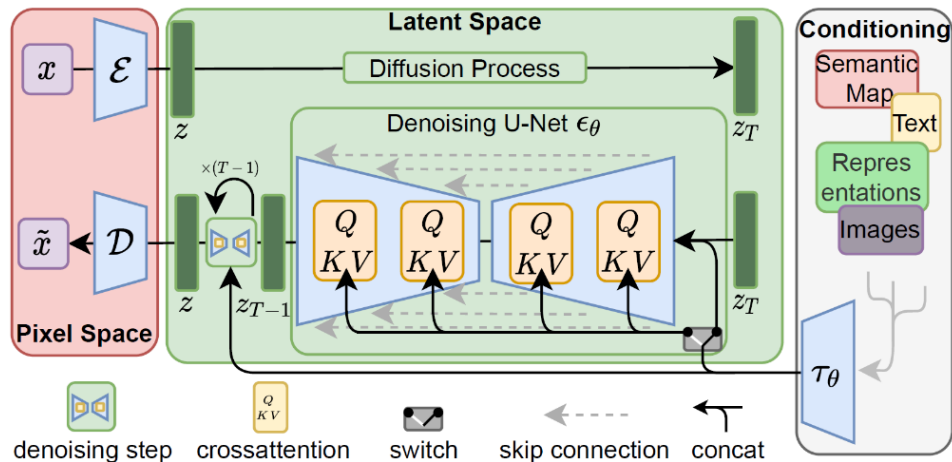
[4] def gen(x):
    q = f"### Question: {x}\n\n### Answer:"
    # print(q)
    gened = model.generate(
        **tokenizer(
            q,
            return_tensors='pt',
            return_token_type_ids=False
        ).to('cuda'),
        max_new_tokens=50,
        early_stopping=True,
        do_sample=True,
        eos_token_id=2,
    )
    print(tokenizer.decode(gened[0]))

[5] gen('Tell me about alpaca?')
```

Image Generation Model

Stable Diffusion

- ▶ <https://stability.ai/blog/stable-diffusion-public-release>
- ▶ AI models trained to restore normal images from noise images
- ▶ Text Encoder (CLIP) converts input text to text embedding.
- ▶ Input random vectors and text embeddings in Unet and generate images by denoising n times.



```
[1] !pip install tensorflow keras_cv --upgrade --quiet
```

```
[2] import time
import keras_cv
import tensorflow as tf
import matplotlib.pyplot as plt
```

```
model = keras_cv.models.StableDiffusion(img_width=512, img_height=512)
```

```
[4] def plot_images(images):
    plt.figure(figsize=(20, 20))
    for i in range(len(images)):
        ax = plt.subplot(1, len(images), i + 1)
        plt.imshow(images[i])
        plt.axis("off")
```

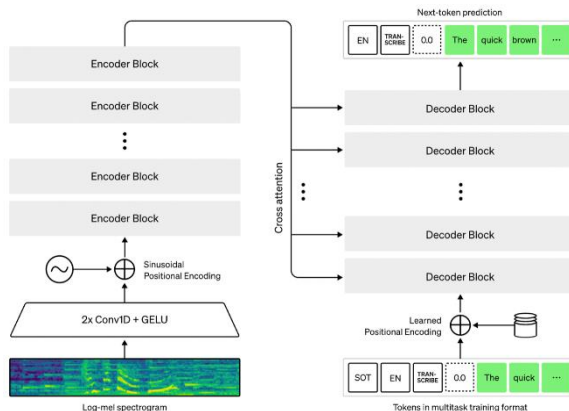
```
[5] images = model.text_to_image("photograph of an astronaut riding a horse",
                                batch_size=3)
    plot_images(images)
```



Automatic Speech Recognition Model

Whisper

- ▶ <https://openai.com/research/whisper>
- ▶ AI models that generate text from voice
- ▶ Whisper trained on 680,000 hours of multilingual and multitask data collected from the web.
- ▶ Use Transformer encoder-decoder architecture.
- ▶ Support Transcribe and Translate.



```
[1] !pip install git+https://github.com/openai/whisper.git
```

```
[2] !wget https://www.openslr.org/resources/12/test-clean.tar.gz
    !tar -zxvf test-clean.tar.gz
```

```
[3] filepath = './LibriSpeech/test-clean/1089/134686/1089-134686-0000.flac'
```

```
[4] from IPython.display import Audio, display
    Audio(filepath)
```

▶ 0:00 / 0:10

```
[5] import whisper
```

```
model = whisper.load_model("base.en")
```

```
[6] result = model.transcribe(filepath, language="en", fp16=False)
    print(result["text"])
```

Text to Speech Model

| Tortoise TTS

- ▶ <https://github.com/neonbjb/tortoise-tts>
- ▶ https://nonint.com/static/tortoise_v2_examples.html
- ▶ AI models that generate voice from text
- ▶ Strong multi-voice capabilities
- ▶ Highly realistic prosody and intonation
- ▶ TTS voice conversion with just a few samples

```
[1] !pip3 install -U scipy

!git clone https://github.com/jnordberg/tortoise-tts.git
!cd tortoise-tts
!pip3 install -r requirements.txt
!pip3 install transformers==4.19.0 einops==0.5.0 rotary_embedding_torch==0.1.5 unicode==1.3.5
!python3 setup.py install

[2] import torch
import torchaudio
import torch.nn as nn
import torch.nn.functional as F

import IPython

from tortoise.api import TextToSpeech
from tortoise.utils.audio import load_audio, load_voice, load_voices

tts = TextToSpeech()

[3] voice_samples, conditioning_latents = load_voice("tom")

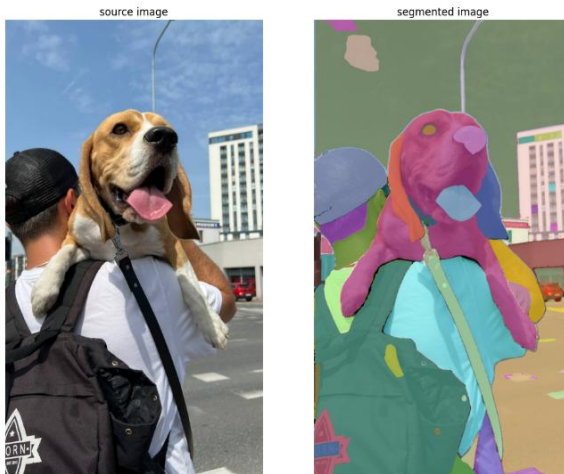
[4] text = """Joining two modalities results in a surprising increase in generalization!
What would happen if we combined them all?"""

gen = tts.tts_with_preset(text, voice_samples=voice_samples, conditioning_latents=conditioning_latents,
                          preset="fast")
torchaudio.save('generated.wav', gen.squeeze(0).cpu(), 24000)
IPython.display.Audio('generated.wav')
```

Image Segmentation Model

Segment Anything

- ▶ <https://github.com/facebookresearch/segment-anything>
- ▶ <https://github.com/facebookresearch/segment-anything>
- ▶ AI model can "cut out" any object in any image.
- ▶ Model can enable text-to-object segmentation.



```
[1] import os
HOME = os.getcwd()
import sys
%cd {HOME}

!(sys.executable) -m pip install 'git+https://github.com/facebookresearch/segment-anything.git'

[2] !pip install -q jupyter_bbox_widget roboflow dataclasses-json supervision

[3] !mkdir {HOME}/data
%cd {HOME}/data
!wget -q https://media.roboflow.com/notebooks/examples/dog.jpeg
!wget -q https://media.roboflow.com/notebooks/examples/dog-2.jpeg
!wget -q https://media.roboflow.com/notebooks/examples/dog-3.jpeg
!wget -q https://media.roboflow.com/notebooks/examples/dog-4.jpeg

[4] !mkdir {HOME}/weights
%cd {HOME}/weights
!wget -q https://dl.fbaipublicfiles.com/segment_anything/sam_vit_h_4b8939.pth

[5] import torch
from segment_anything import sam_model_registry, SamAutomaticMaskGenerator, SamPredictor
DEVICE = torch.device('cuda:0' if torch.cuda.is_available() else 'cpu')
CHECKPOINT_PATH = os.path.join(HOME, "weights", "sam_vit_h_4b8939.pth")
sam = sam_model_registry['vit_h'](checkpoint=CHECKPOINT_PATH).to(device=DEVICE)

[6] mask_generator = SamAutomaticMaskGenerator(sam)

[7] IMAGE_PATH = os.path.join(HOME, "data", "dog.jpeg")

[8] import cv2
import supervision as sv

image_bgr = cv2.imread(IMAGE_PATH)
image_rgb = cv2.cvtColor(image_bgr, cv2.COLOR_BGR2RGB)

sam_result = mask_generator.generate(image_rgb)

[9] mask_annotator = sv.MaskAnnotator()

detections = sv.Detections.from_sam(sam_result=sam_result)
annotated_image = mask_annotator.annotate(scene=image_bgr.copy(), detections=detections)

sv.plot_images_grid(
    images=[image_bgr, annotated_image],
    grid_size=(1, 2),
    titles=['source image', 'segmented image'])
```

Unit 3.

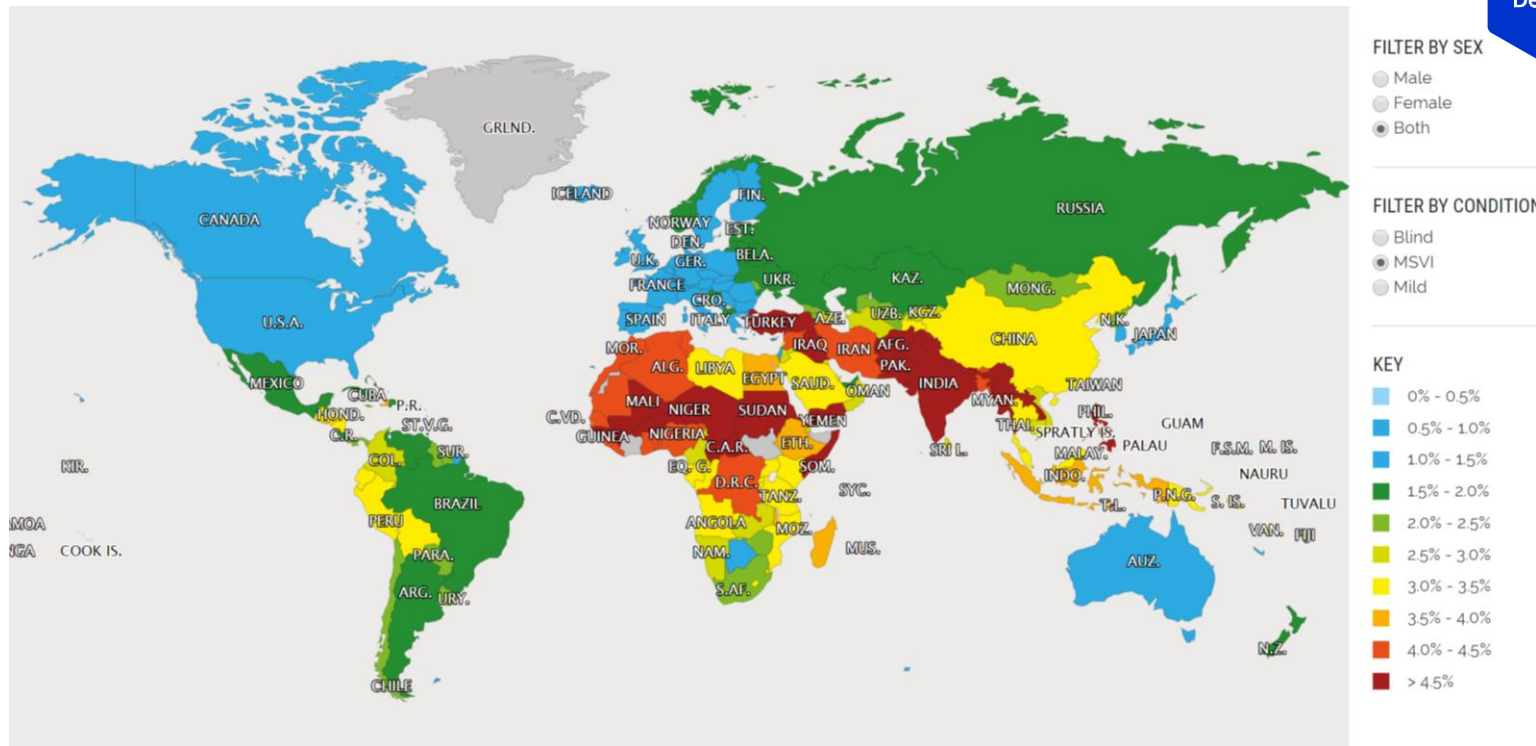
AI Application Cases

| 3.1. AI-based Clinical Decision Support System

| 3.2. Other AI Application Cases

AI-based Clinical Decision Support System

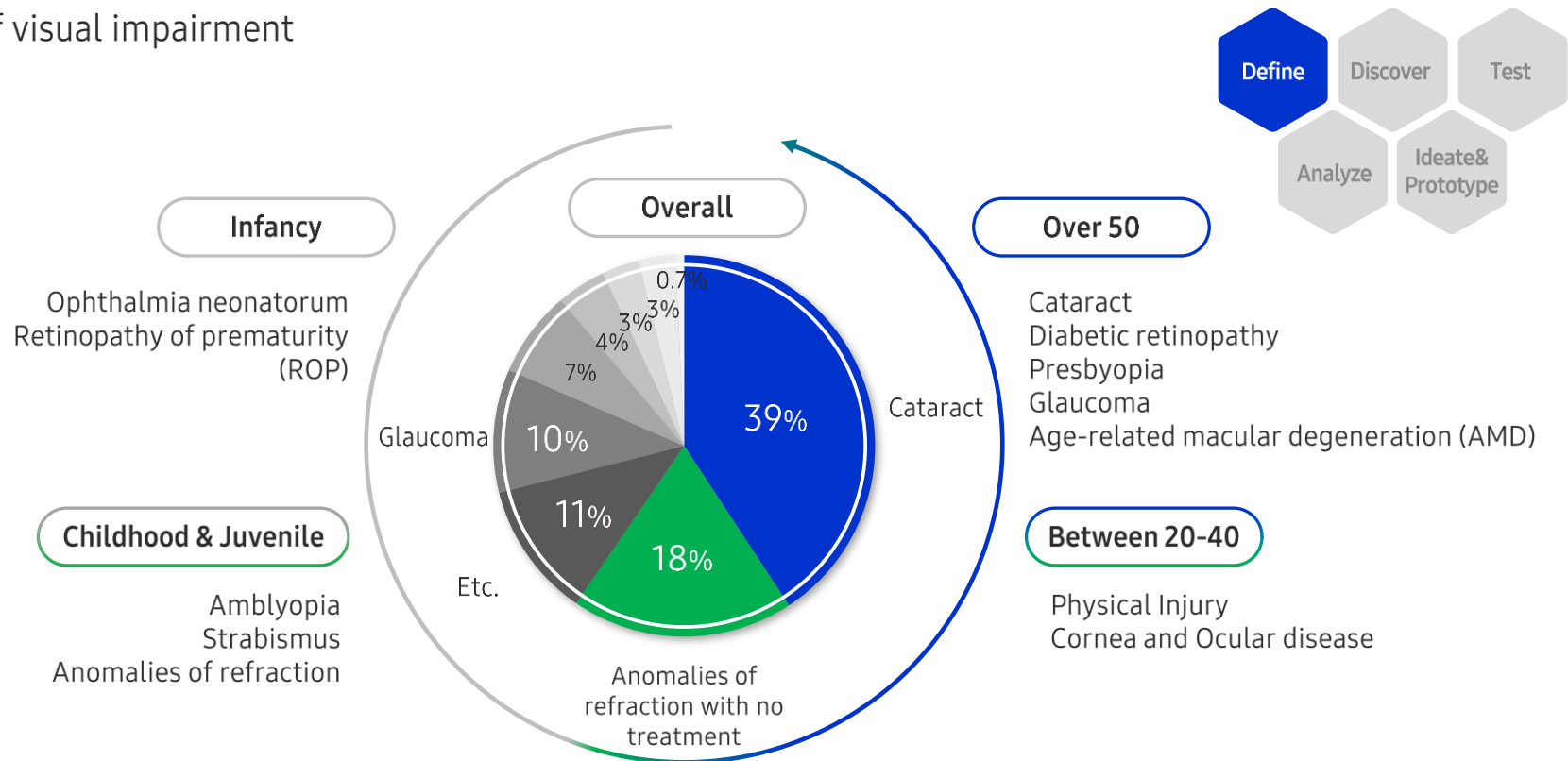
MSVI (Moderate & Severe Visual Impairment) map



Source: IAPB Vision Atlas, Global Vision Database Maps

AI-based Clinical Decision Support System

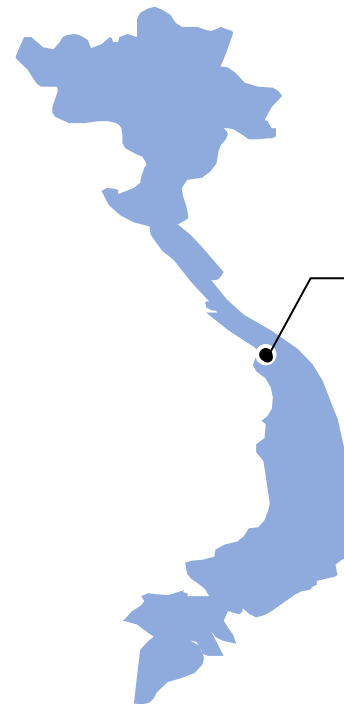
Causes of visual impairment



AI-based Clinical Decision Support System

Vietnam

- ▶ Suitable for initiating ocular healthcare projects due to the high rate of blindness and the asymmetry of accessibility in ocular healthcare by region.
- ▶ High rate of mobile penetration and high-speed internet penetration, and high digital device affordability substantiate the potential of expansion of the project in the country.
- ▶ Of the total population, 2.83% have Moderate & Severe Visual Impairment (In 2015, among 93 million people, 2.5 million were affected).
- ▶ Mid-income level countries can solve cataract problems with their own capabilities. Whereas developing countries cannot afford to handle cataracts, in addition to glaucoma, macular degeneration, and diabetic retinopathy.
- ▶ Especially when it comes to diabetes, every one in three patients with diabetes suffers from diabetic retinopathy.

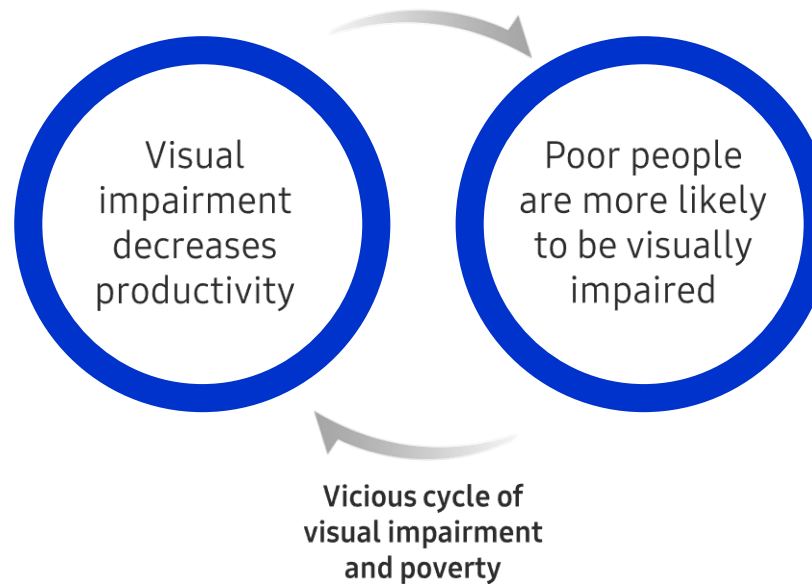
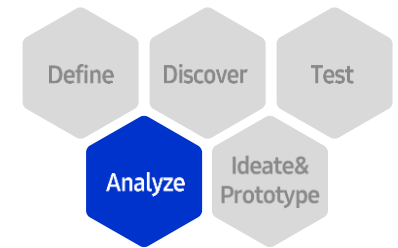


Quang Tri

- The region has the lowest average income among 63 districts in the country.
- Poverty rate is 21.7% in 2015 (national average: 12.5%).
- Low agricultural productivity due to the defoliant sprayed during the Vietnam war.

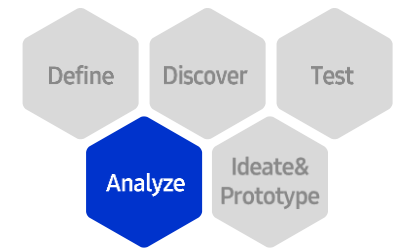
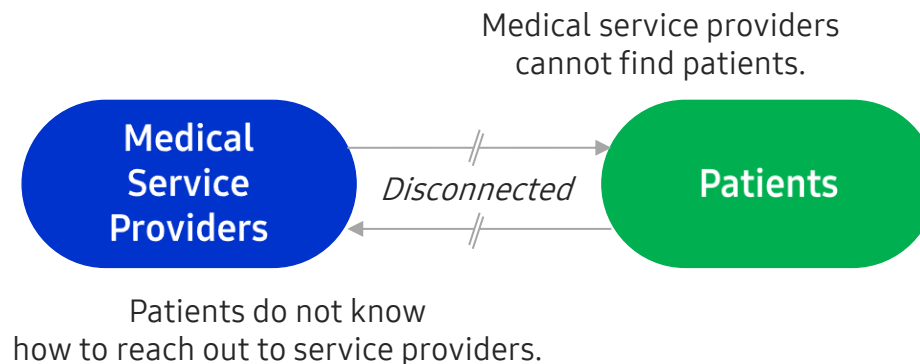
AI-based Clinical Decision Support System

| Why ocular health is important?



AI-based Clinical Decision Support System

Problem definition



- ▶ Most important and difficult thing in preventing visual impairment is the discovery of patients.
- ▶ Patients do not know whether they have the ocular disease or not. Even though they do, they do not know the route to get proper medical treatment.
- ▶ Service providers also cannot locate where to discover untreated patients.
- ▶ Thus, connecting service providers and patients is the most critical activity to break out of the vicious cycle of poverty and visual impairment.

AI-based Clinical Decision Support System

Interview (Patients)

- ▶ “I am blind in one eye. I should visit Hue city, which is far away from here, if I want to get a diagnosis. If the ocular problem can be diagnosed at the commune health center level, local community residents, especially elderly people, will be far more benefitted than now.”
- ▶ “I went to the commune health center because I lost visual acuity gradually. I transferred twice from the commune health center to the local clinic and from the clinic to the provincial hospital because they had no ophthalmoscope to diagnose. If proper diagnostic devices are equipped at the commune health center level, It would be much better to provide health care service.”
- ▶ “I was diagnosed with cataracts at Quảng Trị Provincial Hospital and have not taken the surgery yet. When I first went to the commune health center, they barely helped me. Once an ophthalmoscope is equipped, chronic ocular diseases will be found and treated earlier than now.”



Ms. N (Age: 70,
not a patient)



Mrs. G (Age: 65, Anomalies
of refraction)



Mr. N (Age: 66, not a patient)



AI-based Clinical Decision Support System

Interview (Medical Professionals)

- ▶ “If ophthalmoscopes are distributed to local communities, patients will be transferred to provincial hospitals at the right time, enabling them to get treatment in a timely manner. This will improve the ocular health of patients significantly. There are countless cases where a disease cannot be treated because it has progressed too far.”

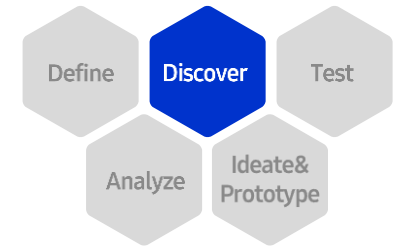


Dr. D, Ophthalmologist
Quảng Trị Provincial Hospital

- ▶ “Most patients come to the clinic or commune health center when they feel pain in the eyes. Before feeling the pain, there is no one to tell them about ocular diseases. If an ophthalmoscope is equipped in each commune health center, it will be more than helpful. Medical staff can diagnose ocular diseases in the early stage, thus making patients transferred to the hospital and preventing visual impairment by providing proper care.”



Dr. V, Head of Center
Commune Health Center



AI-based Clinical Decision Support System

Existing ophthalmoscopes

- ▶ Most critical shortcoming commonly found is that those devices are all made only for ophthalmologists.
- ▶ Making ophthalmic devices portable does not help solve the problem for ocular patients at all.

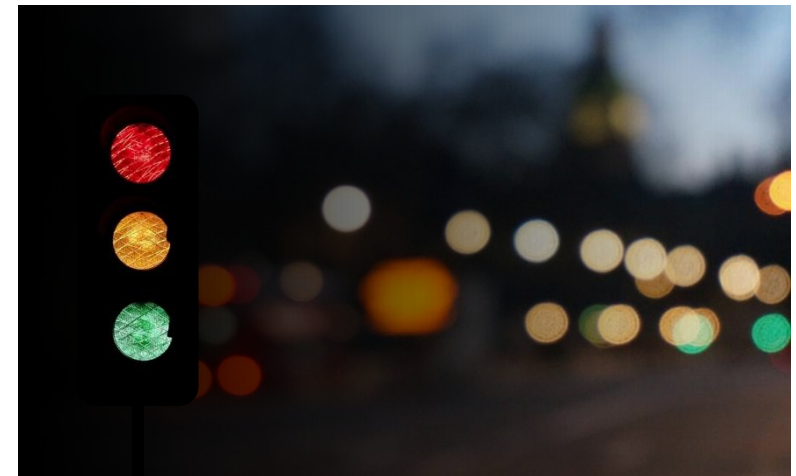
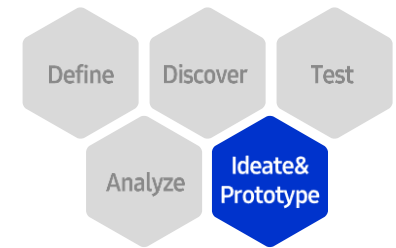


Organization	Research Description	Application
PEEK Vision (UK)	Photograph fundus using smartphones	Get a medical specialist's interpretation of image transmission via message service
Welch Allyn (US)	Take a zoomed-in photograph with portable ophthalmic devices and smartphones	Hard to focus because of the miosis and eye movement by light
Goblink (US)	- Measure visual acuity, anomalies of refraction, and pupillary distance with smartphones - Commercialized using portable ophthalmic devices and smartphones - Deliver information to optometrist and optician	- Direct synchronization with smartphones is unsupported - Cannot obtain information of disease other than anomalies of refraction
Horus (Taiwan)	- Photograph fundus without using mydriatic - Support high-quality image and video - Real-time screen sharing via wi-fi	- High retail price deters mass adoption - Requires a high level of skills for non-specialist doctors
Vula Eye (Switzerland)	- Connect patients living in mountainous areas with medical specialists - Conduct a visual acuity test with a simple guidance - Obtain accurate and detailed information via web chat with specialists	Restricted applicability to cataract and retinal diseases, which are major causes of blindness

AI-based Clinical Decision Support System

| Solution

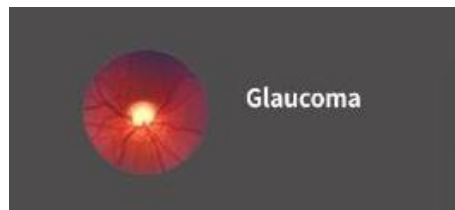
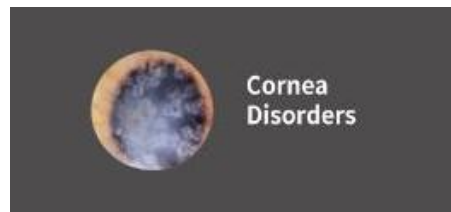
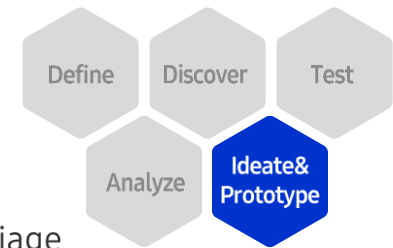
- ▶ EYELIKE: My ocular health monitor with “traffic light”
 - ▶ Alerting with three different colors, the device automatically “Analyzes” whether a medical specialist’s help is required and “Recommends” proper action for each color.
 - ▶ Local medical staff in public healthcare centers can handle the device with “minimum training.”



AI-based Clinical Decision Support System

Pattern Recognition Machine Learning Algorithm

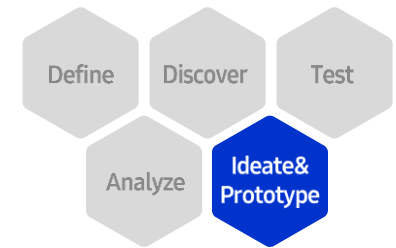
- ▶ Diagnosis support algorithm based on image processing
- ▶ 40,000 images are being accumulated per month
- ▶ Detect anomalies from eye images based on Machine learning and classify the image for triage



AI-based Clinical Decision Support System

Hardware

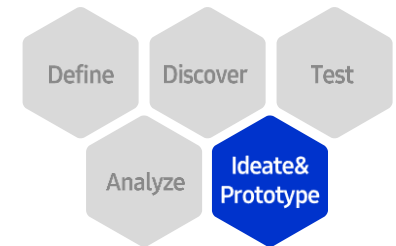
- ▶ “Trash to treasure”
 - ▶ A portable, easy to use, digital ophthalmoscope at an affordable price range utilizing a recycled mobile handset
 - ▶ Key components of the portable ophthalmoscope are an inexpensive yet high-quality camera which is a good fit for mass production and a telecommunication module to connect medical specialists dispatched over a wide range of area
 - ▶ Without ophthalmologist training, a user can easily photograph the fundus with minimum training.



AI-based Clinical Decision Support System

Hardware

- ▶ Living Lab strategy
 - ▶ Distributed EYELIKE digital ophthalmoscopes to 20 commune health centers in Quang Tri province and provided user training.
 - ▶ First, the project team made a prototype, met local health workers and patients, and got feedback.
 - ▶ After spending one more year for discussion, the team created the 2nd prototype that incorporated all stakeholder opinions in the development process, and then it was successfully implemented.
 - ▶ This project will be expanded to another 20 commune health centers in other provinces.



2016



2017



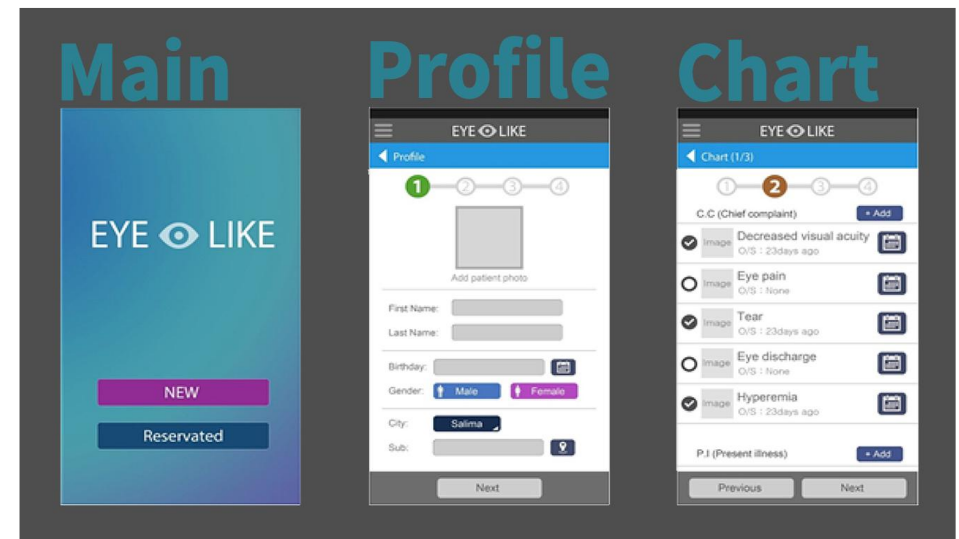
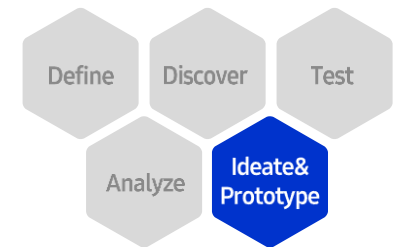
2018



AI-based Clinical Decision Support System

Software

- ▶ Ocular health information management application based on Android OS
 - ▶ A web-based dashboard for the collection, analysis, and visualization of patient information
 - ▶ Distinguish ocular patients
 - ▶ Store lesion images and basic patient information
 - ▶ Locate the residential areas of the patients with GPS coordinates (Creating a disease outbreak map)
 - ▶ Establish an ocular health information system (applicable to the overall public healthcare information system)
 - ▶ Analyze prevalence data



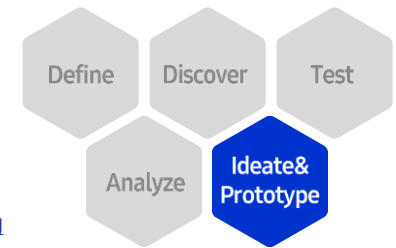
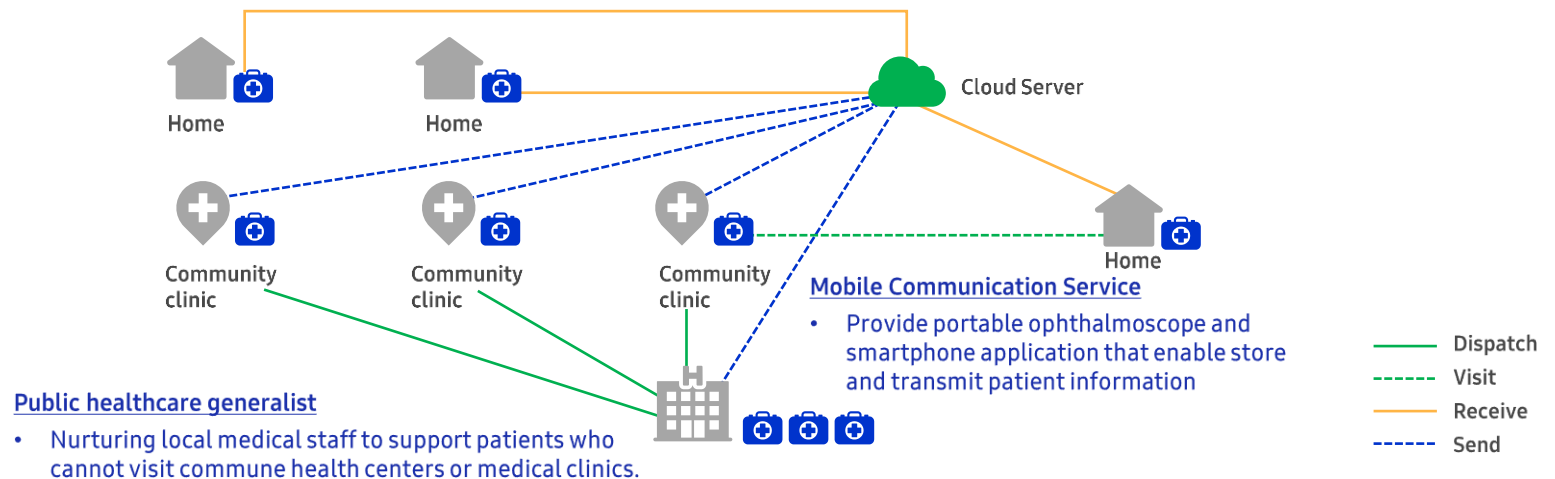
AI-based Clinical Decision Support System

Ocular healthcare service system

- ▶ Connect patients with public healthcare service provider
 - ▶ Utilize commune health centers and their medical staff
 - ▶ Task shifting with EYELIKE solution

Artificial Intelligence based Clinical Decision Supporting System for blindness causing eye diseases

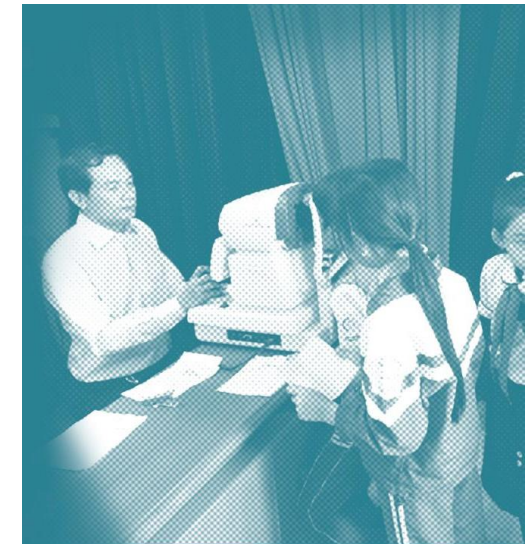
- Improve diagnosis accuracy by providing diagnosis report function based on remote diagnosis algorithm



AI-based Clinical Decision Support System

| Solution Benefit Summary

- ▶ Boost the ocular diagnostic capability of unskilled healthcare staff
- ▶ Create a synergy effect with local ocular healthcare projects which focus on treatment (Enhance efficiency and effectiveness)
- ▶ Reduce the number of ophthalmologist's initial examinations and improve the working environment
- ▶ Reduce ocular healthcare costs from a comprehensive public healthcare perspective
- ▶ Provide portable digital ophthalmoscope with a significantly affordable cost (merely 1% of existing devices' costs)
- ▶ Increase the blind or visually impaired population's productivity
- ▶ Accommodate blind or visually impaired population into the working-age population
- ▶ Improve the quality of life and lower the prevalence of ocular diseases





Unit 3.

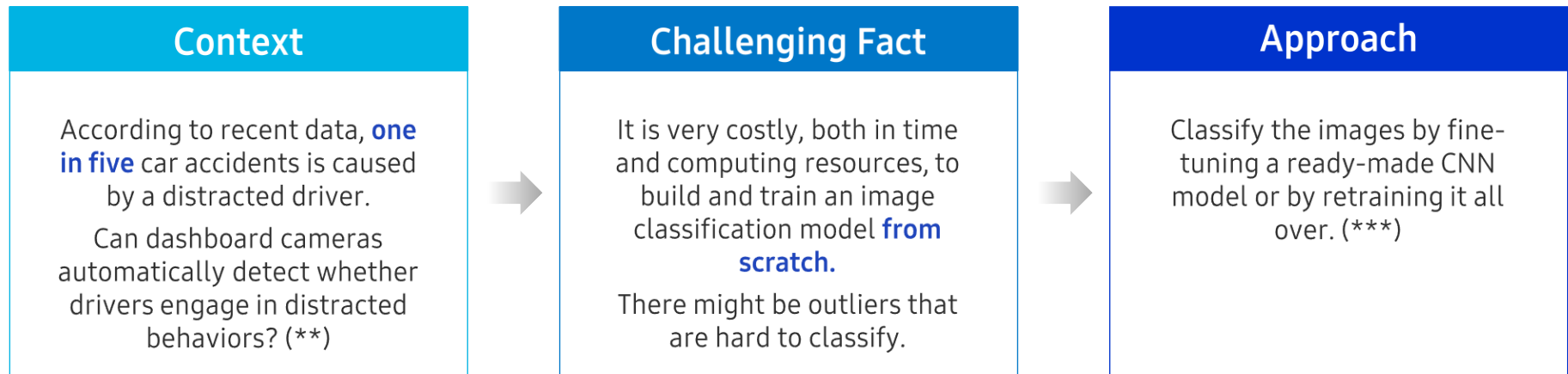
AI Application Cases

| 3.1. AI-based Clinical Decision Support System

| 3.2. Other AI Application Cases

AI Application Cases

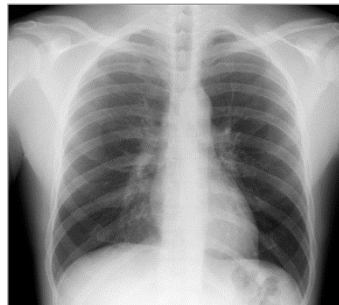
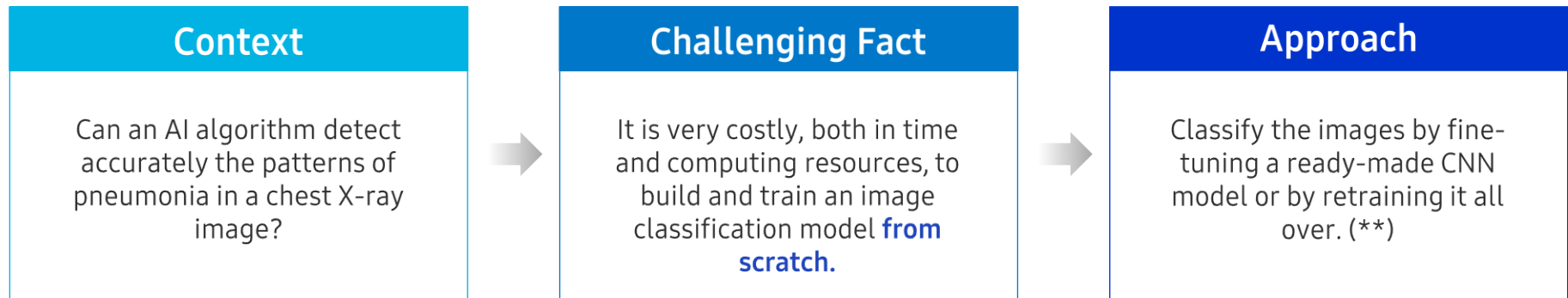
| Case #1: Distracted driver detection



Distracted?

AI Application Cases

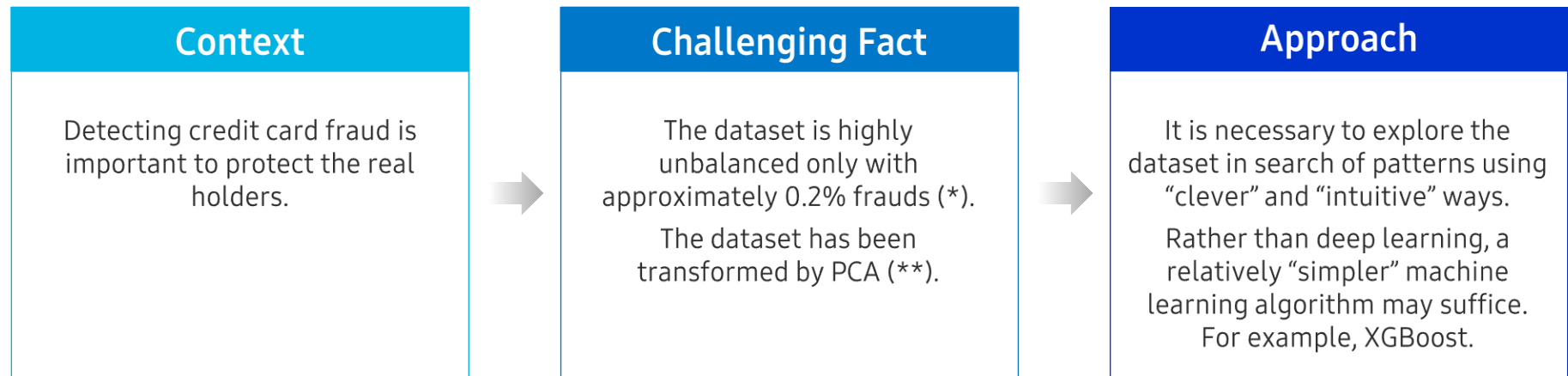
| Case #2: RSNA pneumonia detection (*)



Healthy or else?

AI Application Cases

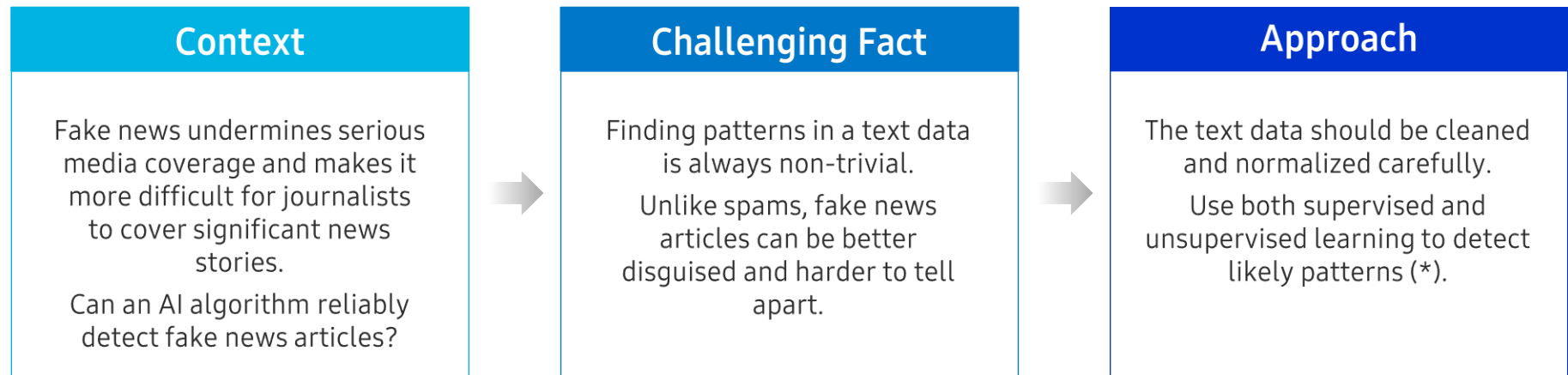
Case #3: Credit card fraud detection



Fraudulent transaction?

AI Application Cases

| Case #4: Identifying fake news



Yet, another fake news?



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